What a dissipative quantum system remembers

Markus Baumann [1,2], Itamar Fink [2], Johannes Wittmann [2], Claudia Linnhoff-Popien [1,2], and Jonas Stein [1,2]

[1]Quantum Applications and Research Laboratory (QAR-Lab), Ludwig-Maximilians-Universität München, Munich, Germany
[2]Mobile and Distributed Systems Group, Ludwig-Maximilians-Universität München, Munich, Germany

Quantum reservoir computing processes a stream of inputs with the natural dynamics of a driven quantum system and trains only a linear readout of measured expectation values. An isolated quantum system never forgets, so its environment decides how long it remembers. We show that the form of the dissipation, set by its jump operators, matters as much as its strength. When each qubit decays on its own, stronger dissipation shortens the memory. When all qubits decay through one shared channel, stronger dissipation lengthens it: interference between the decay paths creates dark states that cannot decay, and the stronger the dissipation, the longer the quantum Zeno effect holds the system there. A long memory has a price: later inputs mix earlier ones into products that a linear readout cannot use. For a broad class of reservoirs, classical or quantum, we prove that the average number of inputs mixed together grows as the product of how strongly one input turns the stored state and how long the memory lasts. The input must therefore weaken as the memory grows, which more than doubles the memory a linear readout recovers from exact expectation values. With finitely many measurements, weak independent decay keeps the most memory, because it cools the qubits towards a state the inputs can rotate. Dark-state memories last just as long but give weaker signals.

Corresponding author
Markus Baumann markus.baumann@campus.lmu.de

1 Introduction

Throw pebbles into a pond, and the ripples hold a fading record of the last few throws. A driven quantum system does the same, and since an isolated quantum system never forgets, its environment decides how long the record lasts [1]. Reservoir computing turns such a record into a machine [2, 3]: a fixed dynamical system, the reservoir, is driven by a stream of inputs, and only a linear readout of its state is trained. In quantum reservoir computing (QRC) the reservoir is quantum [4], here a few interacting spins whose expectation values are read out. Damping, absorption, measurement and resets all create or regulate its fading memory [5–14].

The usual knob is the loss rate. Tuning how fast each qubit relaxes improves a driven spin reservoir [6], memory peaks at an intermediate decay that coincides with maximal optical absorption [8], and drive, interactions and damping have been tuned together [15], also by us [16]. In this picture dissipation is a dial with a sweet spot: the more the reservoir leaks, the faster it forgets.

But dissipation can be designed, not just dialled. Its jump operators describe how a reservoir leaks: each spin into its own channel, or all spins through one shared channel. In a shared channel the decay paths interfere, and some collective states, the dark states, cannot decay at all, as in subradiance [17–19]. Dissipation engineering chooses jump operators to prepare, protect and even compute with quantum states [20–23]. Earlier QRC studies tuned local loss rates [6], compared local noise types [24, 25] or photon loss in Kerr oscillators [26], and read a static classifier out of a decoherence-free subspace [27]. Here we ask how the shape of the dissipation, not only its strength, sets the memory of a quantum reservoir.

We find three things, which become three design rules (Fig. 1). First, the jump operators decide

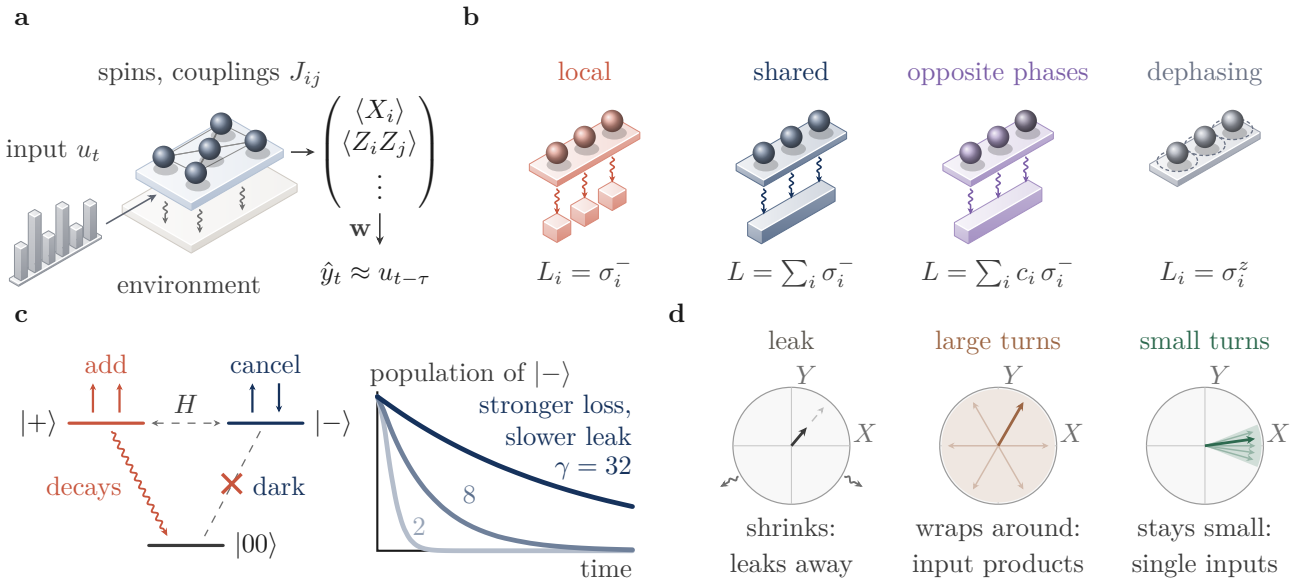


Figure 1: **Engineering how a quantum reservoir leaks. a** Inputs drive interacting spins, and a trained linear readout recalls past inputs from the spins' expectation values. **b** Four shapes of the environment at equal total coupling. With opposite phases the darker spins couple to the shared channel with the opposite sign. **c** Two spins sharing one channel: the decays of $|10\rangle$ and $|01\rangle$ add for $|+\rangle$ and cancel for $|-\rangle$, which is dark. The Hamiltonian lets $|-\rangle$ leak through $|+\rangle$, and the stronger the loss, the slower the leak (inset). **d** Two ways to forget: a stored input leaks away, or later inputs turn the stored state at random until only products of inputs remain. Small turns keep each input readable.

whether loss shortens or lengthens the memory time, the clock of the reservoir. At the same total coupling, spins that leak on their own forget faster as the loss grows, while spins that share one channel forget more slowly, because strong loss holds the reservoir in its dark states through the quantum Zeno effect [28–30]. Second, the input must follow this clock. Each later input turns the stored state by a small, input-dependent angle. As the turns add up, an old input survives only multiplied by the inputs that followed, in products such as $u_{t-3}u_{t-5}$ that a linear readout cannot use. For reservoirs in which a stored signal fades along a single slow direction, we prove that the number of inputs mixed into a memory is exponentially distributed, with mean equal to how strongly one input turns it times how long it lasts. Nothing in this law is quantum. Dissipation enters through its two factors. An input that weakens as the memory grows therefore more than doubles the linear memory with exact expectation values. Third, the shape decides how large, and hence how measurable, a memory is. Once measurements are counted, weak local loss keeps the most memory by cooling the spins, and dark-state memories last as long but are fainter. Dark states pay where a reservoir must dissipate strongly: a dark-state memory of T steps needs collective loss about $4T^2$ times stronger than ordinary local loss, and today's waveguide dark states [31] reach about five steps.

Some ingredients are known. Classical networks have known limits on short-term memory [3, 32, 33], and a stronger input trades memory for nonlinearity [34, 35], also across quantum reservoir protocols [36]. Weakening the input as the memory grows has a classical counterpart in leaky-integrator echo-state networks [37]. States protected by the Zeno effect are known to leak slowly [29, 38]. Three results are new. Under a shared channel, loss lengthens the memory, which turns Zeno protection into a memory resource. The channel phases, the signs with which the spins couple to the channel, decide whether the input reaches the dark states. The mixing law ties the input to the clock. The law is proved for a class of reservoirs (Appendix B) and the three-spin limits are certified by interval arithmetic (Appendix D). Together, these results make the jump operators a design variable for memory: they decide how long a quantum reservoir remembers, how strongly it may be driven, and how much of its memory survives measurement.

2 Model

We test these ideas on the driven spin reservoir of Ref. [6]. Five spins ($N = 5$) interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with Pauli operators X_i, Z_i of spin i and a field strength h. The environment acts through jump operators L_k in a Lindblad master equation [39, 40],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2} \{ L_k^\dagger L_k, \rho \} \Big), \tag{2}$$

with all L_k scaled by one loss rate γ. A linear readout of 45 features, the single-spin and two-spin expectation values, is trained to estimate the input from τ steps back. Its recall C_τ, between 0 and 1, is the squared correlation between the estimate and the input on held-out data. The memory $\sum_\tau C_\tau$ adds up the recall over all delays. One unit is one past input recalled perfectly, and the ceiling is 45, the number of features. The total capacity also counts products of past inputs. The memory time t_{mem}, how long the variance of a stored signal survives the random inputs, is the clock of the reservoir (Fig. 3c). Secs. 3.1–3.3 use exact expectation values, as if every measurement were repeated infinitely often, while Sec. 3.4 counts measurements and decides which memories stay readable.

We call the set of jump operators the shape of the environment and γ its strength. Every shape gets the same total coupling $\mathrm{Tr}\, K = \sum_k \mathrm{Tr}\, L_k^\dagger L_k$, which fixes the mean decay rate over all 4^N modes of the reservoir, for every Hamiltonian (Appendix A.2). It does not fix the slow modes, which can become slower or faster, so Sec. 3.4 also compares the shapes at a matched clock. Of the eight shapes in Table 1, four carry the story (Fig. 1b). With local loss every spin loses its excitation on its own. With one shared channel all spins lose it together, either with equal phases or with opposite phases on spins 4 and 5. With dephasing every spin loses its phase. Dephasing randomizes without favouring any state: it is unital, so every step keeps the fully mixed state fixed whatever the input, and it stores nothing.

Table 1: Shapes of the environment at five spins, each with $\mathrm{Tr}\, K = 80\gamma$, and the number of operators that the dissipator leaves unchanged. The operators $\sigma_i^\mp$ lower and raise spin i, with $i < j$ for pairwise loss, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

shape	jump operators	protected
local	$\sqrt{\gamma}\,\sigma_i^-$	1
shared	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i \sigma_i^-$	100
two channels	$\sqrt{\gamma}\sum_{i\le 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,Z_i$	32

What tells the shapes apart is what the environment cannot reach. Local loss leaves only the ground state alone. A shared channel, with equal or opposite phases, cannot empty ten dark states and leaves every operator on them, 100 in all, unchanged, while dephasing keeps the 32 populations of the spin basis but erases every coherence.

3 Results

3.1 The shape decides whether loss shortens or lengthens the memory

At weak loss all shapes behave alike, but as γ grows they split in opposite directions (Fig. 2). With local loss, recall retreats to ever shorter delays and the memory falls steadily (Fig. 2a,c). With one shared channel the memory first dips and then doubles, and at strong loss inputs fifty steps back are still recalled (Fig. 2b,c). The other shapes split the same way: exchange and gain/loss follow local loss, two channels the shared one, and pairwise loss opposite phases (Fig. 2c, grey). The decay rates show the split (Fig. 2d): without protected operators the leak grows in proportion to γ and the memory shortens, while with dark states the leak falls as $1/\gamma$ and the memory lengthens.

Why does more loss make a shared channel forget more slowly? Take two spins that share one channel (Fig. 1c). Their decay amplitudes cancel for the dark state $|-\rangle$, which can leak only after the Hamiltonian turns it into the bright state $|+\rangle$. A strong channel acts like constant checking of whether an excitation has left, and such checking holds the state where the answer is no: the quantum Zeno effect [28, 41]. The leak

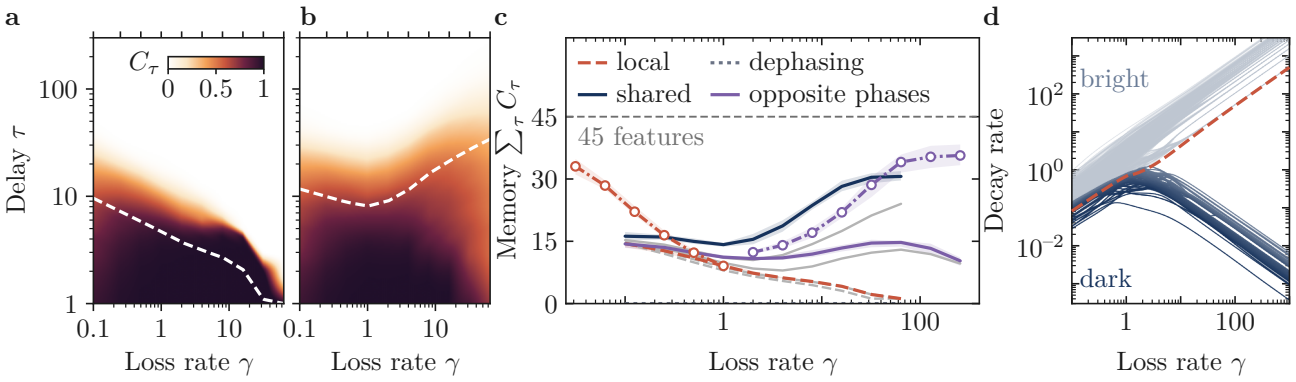


Figure 2: **Same total coupling, opposite clocks. a,b** Recall of the input τ steps back (colour) against the loss rate, with local loss (a) and one shared channel (b). Dashed: the delay that splits the memory in half. More loss shortens the memory in a and lengthens it in b. **c** Memory at the full input strength, the shapes of Fig. 1b in colour and the others in grey. Circles: an input that weakens as the memory grows (Sec. 3.3). **d** Every decay rate of a reservoir with a shared channel: the dark modes slow down as the loss grows, while the slowest mode of local loss (dashed) speeds up.

then falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, just as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

The same holds for any number of spins. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state but cannot make it forget. Forgetting enters only at the next order, when the Hamiltonian briefly lifts a dark state into a bright one that decays [29, 38]. One step is then a rotation plus a leak that falls as $1/\gamma$ (Eq. (A2)): a nearly orthogonal map at the edge of stability, the classic recipe for a long memory in a network [32]. The size of the reservoir sets only the strength of the leak (Appendix E). Local loss has no such regime: at strong loss it protects only the ground state, and a single state cannot hold a record.

Two tests show that the dark states hold the memory (Appendix E). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir at strong loss. Erasing the dark coordinates after every step destroys all memory beyond a few steps, while erasing a rotated copy of the same size leaves most of it.

But lasting is not the same as staying readable. With opposite phases the dark states are just as protected, yet the memory climbs back only to its weak-loss level (Fig. 2c): something besides the leak is erasing it.

3.2 The price of a long memory

That something is the input itself. Each input turns the stored state slightly, by an angle that depends on the input (Fig. 1d): to an old input, every newer input is noise. Over the memory time the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, the record of an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs. We say that later inputs scramble the stored input.

A single input is thus lost in two ways: it leaks away, and it is scrambled. One input stays linearly readable for the time to forget t_{f}, whose rate is the leak plus a scrambling rate that grows as η^2, the squared input amplitude (Appendix A.3). The leak grows as γ with local loss and falls as $1/\gamma$ with dark states. So $\eta^2 t_{\mathrm{f}}$ should depend on loss and input only through γ/η^2 with local loss and through $\gamma\eta^2$ with opposite phases. Both collapses hold for all four input strengths and run in opposite directions (Fig. 3a), so leak and scrambling add. With exact readout, then, no loss rate is best on its own: weakening the input moves the best loss rate (Appendix E).

Here the phases of the channel become a design choice. The dark states are annihilated by the collective lowering operator $\sum_i c_i \sigma_i^-$. An input field $\sum_i c_i X_i$ with the same phases is that operator plus its adjoint, so it cannot couple two dark states. The change that one input makes on the dark states, its write, then comes only from brief excursions into bright states and falls as $1/\gamma$. An input pattern across the phases writes into the dark states directly, and its effect falls only slowly (Fig. 3b). With equal phases the uniform input of Eq. (1) is of the first kind, and its time to forget keeps growing in proportion to γ. With opposite

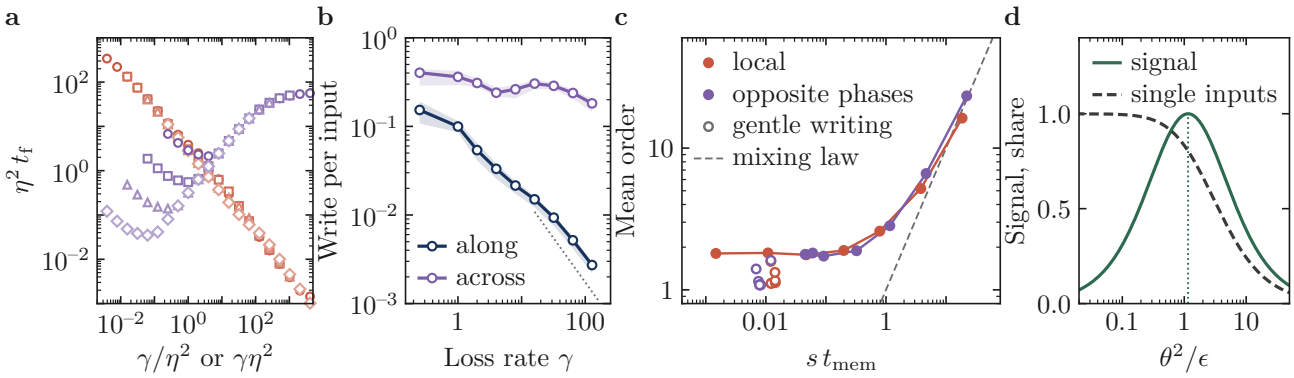


Figure 3: **The price of a long memory, and how the input pays it. a** Time to forget for four input strengths (markers), rescaled as Sec. 3.2 predicts: for each shape all strengths fall on one curve. **b** How strongly one input writes into the dark states of the equal-phase channel: an input along the phases of the channel fades as $1/\gamma$ (dotted), one across them keeps writing. **c** Inputs mixed into the five-spin memory against the mixing law (dashed): at the full input strength they follow it, and gentle writing keeps them near one. **d** One qubit: the stored signal is largest at an intermediate squared turn over a memory time (dotted), where single inputs still carry most of it.

phases it is of the second kind, and its time to forget levels off. The rule is sharp: a phase error δ couples dark states directly at order δ, so beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the one with opposite phases.

How many inputs does a scrambled memory mix? We call the number of distinct inputs that a stored function involves its order: u_{t-3} alone has order one, $u_{t-3}u_{t-5}$ order two, which splits the information processing capacity [34, 42] by inputs. The simplest reservoir with a long memory is reset to a fixed state with a small probability p in each step and otherwise turned by an input-dependent unitary. A stored signal then lasts about $t_{\mathrm{mem}} = 1/(2p)$ steps. How strongly one input turns what is stored is measured by its mixing fraction s, between 0 and 1: the part of a stored signal that one input turns differently from the average input (Appendix B). For small turns it grows as the square of the turn. Our main theorem says that a memory mixes on average $s\,t_{\mathrm{mem}}$ inputs, how strongly one input turns it times how long it lasts:

Theorem 1 (Mixing law, made precise in Appendix B)**.** Let a reservoir be reset with probability p per step and otherwise turned by input-dependent unitaries that mix the stored state, leave a single slow direction of the stored covariance, and write a signal (conditions in Appendix B). Pick a stored function with probability equal to its share of the capacity, and let D be its order. As $p \to 0$, D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

The reason is a race: in each step a fraction s of the stored signal picks up one more input, while a fraction $1/t_{\mathrm{mem}}$ of it is lost, so the number of inputs follows a geometric distribution, which becomes exponential for long memories. The law holds for physical channels without a reset as well, with the leak in place of the reset (Appendix B.5). What it needs is a single slow direction of the stored covariance, not a single slow mode. Nothing in the law is quantum: classical state-affine reservoirs [43] obey it under the same conditions, and it says how the information processing capacity [34] spreads over the orders. Dissipation enters through its two factors, the clock and how hard an input writes.

In the five-spin reservoir Eq. (A3) gives the order distribution exactly. One input turns the stored state only slightly, $s \approx 0.02$ at the full amplitude. As the memory grows, the mean order with opposite phases rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, as the mixing law predicts: it equals $s\,t_{\mathrm{mem}}$ plus an offset of one to two inputs, as Theorem 1 allows (Fig. 3c), and gentle writing (Sec. 3.3) keeps it near one. Mixing is also a resource: products of inputs are the nonlinear memory that tasks such as parity need [34, 35]. At a fixed input strength, though, a longer memory is a more scrambled one.

3.3 The input must follow the clock

How much gentler must the input get? One qubit shows the trade-off in closed form (Fig. 3d, Appendix C). It relaxes towards $+X$ at a small rate ϵ, each input turns it about Z by $\pm\theta$, and it is read along Y. Everything depends on θ^2/ϵ, the squared turn accumulated over a memory time. When it is large, later inputs scramble earlier ones. When it is small, each input writes too little. The signal is largest in between, at $\theta^2/\epsilon = 2/\sqrt{3}$, and a turn that falls exactly as $\epsilon^{1/2}$ is the only power law that keeps the signal finite, since it holds θ^2/ϵ fixed. We call an input amplitude that follows the clock in this way, falling as one over the square root of the memory time, gentle writing: it keeps $s\,t_{\mathrm{mem}}$ fixed as the memory grows.

The five spins follow the same rule in both cases: the amplitude falls as $1/\sqrt{\gamma}$ for the shape whose memory grows with the loss, and as $\sqrt{\gamma}$ for the one whose memory grows as the loss weakens (Fig. 2c, circles). With exact readout this more than doubles the linear memory, from about 15 to 35 units for both shapes (Table 2), and with enough measurements it pays as well (Sec. 3.4). It needs both knobs: weakening the input alone, or tuning the loss alone, falls short. Only the equal-phase channel comes close at the full amplitude, because its uniform input already writes gently.

The gain moves capacity rather than creating it. With exact readout all capacities of a readout add up to its rank, 45 here [34], so raising the linear memory from 15 to 35 units leaves at most 10 instead of 30 for products of inputs. A long memory therefore needs no dark states: weak local loss with gentle writing reaches one too. What dark states add is a memory that stays long under strong dissipation and grows with the loss.

For three spins with one shared channel of opposite phases we prove both regimes (Appendix D): at a fixed amplitude single inputs lose their share, while with an amplitude $\eta_0/\sqrt{\gamma}$ they keep it and are recalled ever further back.

The memory is written into coherences between dark energy levels. Coherence shapes the performance of spin reservoirs [44], and erasing these coherences after every input collapses the stored signal as $1/\gamma^2$ (Appendix D). A gentler input writes a fainter record. With exact readout this costs nothing, but with measurements it has a price, and it reorders the shapes.

Table 2: Linear memory $\sum_\tau C_\tau$ of the five-spin reservoir in units (ceiling 45), mean of eight reservoirs, with exact readout or with a measurement budget of 10^9 reservoir steps per output step (Sec. 3.4).

	local loss	opposite phases	equal phases
tuned loss, $\eta = 1$	14.5	14.7	31[a]
weak input[b]	16	25	–
gentle writing	35 ($\gamma = \frac{1}{64}$)	36 ($\gamma = 256$)	–[c]
10^9, $\eta = 1$	13 ($\gamma \approx \frac{1}{8}$)	12 ($\gamma \approx 16$)	13 ($\gamma = 1$)[d]
10^9, gentle writing	34 ($\gamma = \frac{1}{64}$)[d]	21 ($\gamma \approx 64$)	–[c]

[a] At strong loss. [b] At $\gamma = 1$ and $\eta = \frac{1}{32}$, two reservoirs. [c] The uniform input already writes gently in this channel. [d] Smallest loss rate sampled. The memory falls as the loss grows.

3.4 What survives measurement: cool, don't randomize

Exact expectation values flatter a long memory. A memory that follows the clock grows longer, not larger: it cannot exceed the 45 features, so each delay is recalled more faintly. With local loss at $\gamma = 64$ the previous input moves the state by about 10^{-9}: perfect recall with exact readout, almost none with 10^8 preparations per step.

Finite sampling limits what any physical reservoir resolves [45]. A quantum measurement gives a random outcome, so every feature is an average over n preparations, repeated runs of the reservoir, per measurement setting and step, with all spins read along X, Y or Z, and the regularization of the readout is chosen on held-out data (Appendix A.5). A measurement destroys the state, so each preparation must also replay the inputs that the memory still holds [12], and their number grows with the clock. A fair comparison therefore fixes the budget, the number of steps the reservoir runs per output step: three settings, each with n preparations that replay the last τ_{90} inputs from the fixed state $|00000\rangle$, where the first τ_{90} delays hold 90% of the exact recall. We simulate this replay.

At a fixed budget the loss rate has an optimum again, and the budget sets it (Fig. 4a). At 10^9 preparations times replayed steps, weak local loss with gentle writing keeps the most, 34 units, nearly all of its exact memory, against 21 for opposite phases at their optimum near $\gamma = 64$ (Table 2). Gentle writing pays once each setting is prepared about 2×10^4 times per step with local loss and 10^5 times with opposite phases: the dark-

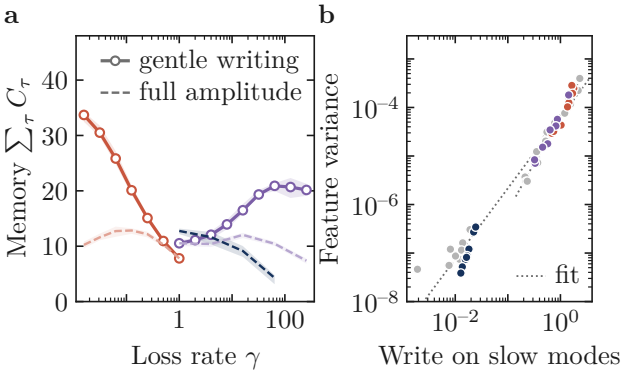


Figure 4: **What survives measurement, and why.** Colours as in Fig. 2c. **a** Memory when each output step may use a fixed budget of measurements (Sec. 3.4): weak local loss with gentle writing keeps the most. **b** The signal grows with the write, how strongly an input moves the state that the dissipation leaves behind, for all shapes at a matched clock. Dotted: power-law fits.

state memory, though as long as that of weak local loss, is three times fainter. The equal-phase channel, whose exact memory grows with the loss, now loses memory as the loss grows.

The size of the signal is set by the state that the dissipation leaves behind. An input acts on the stationary state ρ_* through the commutator with its Hamiltonian, and only the part of this change that lies on the slow modes is remembered: the write onto the slow modes. At one matched clock (Appendix A.5) and input amplitude, the feature variance of the seven shapes that write follows this write over four orders of magnitude (Fig. 4b). Shared channels leave a polarized state that the uniform input moves only through bright states, and they store almost nothing. Local and pairwise loss cool the spins into a state that the input moves directly, and they store the most. Dephasing leaves the fully mixed state and stores nothing. The replay shows the same: restarting every run from the pure state $|00000\rangle$ raises the signal sevenfold with opposite phases, but less than twofold with local loss. With earlier findings that non-unital damping helps and unital noise hurts [7, 46], this gives the third design rule: a reservoir should forget by cooling, towards a low-entropy state, not by randomizing.

4 Conclusion

An isolated quantum system never forgets. A dissipative one remembers only as long as its environment lets it. We have shown that the environment decides far more than how long. The shape of the dissipation, set by the jump operators, decides whether loss shortens or lengthens the memory, how strongly an input may write it, and how much of it survives measurement. Dissipation is therefore not a dial with a fixed sweet spot: the damping sweet spot found at a fixed input scaling [6] moves as the input weakens, and with finite sampling the measurement budget sets it.

First, the jump operators reverse the clock. At the same total coupling, more loss makes spins that decay on their own forget faster, but spins that share one channel forget more slowly: their decay paths interfere into dark states, and the quantum Zeno effect turns stronger loss into stronger protection. Second, a long memory has a price, and we have derived it exactly: for every reservoir, classical or quantum, whose stored signal fades along one slow direction, a memory mixes on average as many inputs as the mixing fraction of one input times the memory time. An input that weakens as the memory grows pays this price and more than doubles the linear memory, with weak local loss even under measurement. Third, measurement decides which memory is usable: dissipation that cools the spins into a state the input can move keeps the most, while dark-state memories last as long but are fainter. Three design rules follow:

(1) Choose the jump operators, not only the loss rate: they decide whether stronger dissipation shortens or lengthens the memory, and their phases decide whether an input reaches the protected states.

(2) Let the input follow the clock: weaken it as the memory grows, so that a stored input mixes with few later ones and stays readable on its own. Once measurements are counted, this pays as soon as each setting is sampled often enough.

(3) Forget by cooling, not by randomizing: let the dissipation drive the system towards a polarized state that the input can move, not towards the fully mixed state.

What is quantum is the protection: the dissipator protects every operator on the dark states, coherences included, the memory is written into coherences between dark energy levels, and the phases of the channel decide whether an input can

reach it. The direction of the clock, the phase rule and the mixing law need only the protected operators, the input operator and the conditions of Appendix B, not the details of the couplings. They therefore carry over to any platform that shares a decay channel among its qubits: shared resonators, waveguides and structured couplers [47–50]. In a waveguide the collective decay comes with the coupling, the positions of the spins set the phases [31], and the bath is flat enough for a Markovian description up to $\gamma \approx 64$ at couplings of order a megahertz [49]. A dark-state memory of T steps needs collective loss about $4T^2$ times stronger than local loss (Appendix E), which sets the target for waveguide experiments.

How a quantum system leaks decides not only how fast it forgets, but what it keeps and in what form.

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Data availability

The code, the data behind every figure and the interval certificates of the three-spin limits are available at `https://github.com/eybmits/what-a-dissipative-quantum-system-remembers`, and the certificates are also attached to this PDF.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing, I.F. and J.W. to methodology, interpretation and editing, and C.L.-P. and J.S. to supervision and conceptual development.

A Methods

This appendix specifies the five-spin simulations and defines the quantities that the main text uses: the reservoir and its readout, the equal total coupling of the shapes, the slow modes and the strong-loss generator, the order of a memory and the measurement protocol.

A.1 Reservoir, readout and memory

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, and every shape uses the same eight seeds, and runs with the input on one spin use the first four. After a washout of $\max(800, 60\gamma)$ steps, at least four times the time to forget, we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. The memory sums C_τ over delays 1 to 300. A delay without any recall reads about $1/1600$ on the test inputs, so the sum has a floor of about 0.2 units. Figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile.

A.2 Shapes and equal coupling

Table 1 lists the jump operators. Their weights give every shape the same total coupling $\operatorname{Tr} K = N 2^{N-1}\gamma = 80\gamma$, which fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the generator obey

$$\frac{1}{4^N} \sum \operatorname{Re} \lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{A1}$$

for every shape and every Hamiltonian. The protected operators form the kernel of the dissipator.

A.3 Slow modes, time to forget and the strong-loss generator

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the dissipator protects n operators, the slow modes are the $n-1$ slowest decaying ones besides the stationary state, and the time to forget is $t_\mathrm{f} = 1/(\Delta t\,\bar{\kappa})$, with $\bar{\kappa}$ their mean rate. Without protected operators we use the slowest mode. The mean rate adds the leak and the scrambling exactly (Appendix E). Figure 2d shows instead the decay rates $-\operatorname{Re}\lambda$ of the generator at the mean input, $u = 1/2$. With a shared channel we call the $n-1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, eliminating the decaying operators to second order [29, 38] gives the generator of the protected coordinates,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \qquad \text{(A2)}$$

with $\mathcal{H} = -i[H,\cdot\,]$. The first term is the Hamiltonian restricted to the protected operators, which turns them, and the second is the leak through brief excursions into bright states.

For gentle writing we replace $h(1+u)$ in Eq. (1) by $h[\frac{3}{2} + \eta(u-\frac{1}{2})]$, which keeps the mean field: $\eta = \min(1, \gamma^{-1/2})$ for opposite phases and $\eta = \min(1, \gamma^{1/2})$ for local loss.

A.4 Order of a memory

A stored signal can depend on one past input or on products of several, and its order counts them: u_{t-3} has order one, $u_{t-3}u_{t-5}$ order two. Precisely, for independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs. The order of a part is the size of its set. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$, and for continuous inputs one uses orthogonal polynomials in each input. Quantum channels act affinely on traceless coordinates [51, 52], $r_t = A(u_t)r_{t-1} + b(u_t)$. Let $\bar{A} = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the change that one input writes, with covariance V. With $\mathcal{B}X = \bar{A}X\bar{A}^\mathsf{T}$, $\mathcal{T}X = \mathbb{E}[AXA^\mathsf{T}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k. \qquad \text{(A3)}$$

The innovation adds the current input alone. The mean step carries a stored product along without adding an input. The fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the stationary covariance Σ. A readout of features $y = Mr$ explains the share $\operatorname{Tr}[(M\Sigma M^\mathsf{T})^{+}MG_kM^\mathsf{T}]$ of order k, and these shares add up to the rank of the readout [34]. For one observable with coordinates a the share of order k is $a^\mathsf{T}G_ka/a^\mathsf{T}\Sigma a$, and the share of single past inputs is $a^\mathsf{T}\bar{A}G_1\bar{A}^\mathsf{T}a/a^\mathsf{T}\Sigma a$.

The memory time and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1 - 1/t_\mathrm{mem}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s = \ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix B.5. For five spins we evaluate Eq. (A3) on the 1023 traceless Pauli coordinates (Appendix E).

A.5 Finite sampling

For each step we form the outcome distributions of the three settings and draw n preparations from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. In Sec. 3.4 the distributions come from the replay: each output step replays its last τ_{90} inputs from $|00000\rangle$, with $n = 10^3$ to $10^{8.5}$ in half decades or, for a budget B of preparations times replayed steps per output step, $n = B/(3\tau_{90})$. Only Fig. 4b uses the state with the full history. The matched clock of Fig. 4b is a leak time $t_\mathrm{leak} = 1/(\Delta t\,\bar{\kappa}_0)$ of 32 steps, where $\bar{\kappa}_0$ is the mean decay rate of the slow modes of the generator at the mean input (Appendix E).

B The mixing law

This appendix proves the mixing law of Sec. 3.2: in a reservoir that forgets slowly, a stored function involves on average as many inputs as the mixing fraction of one input times the memory time. The last part extends the law to physical channels without a reset, which Appendix D needs.

B.1 Setting and statement

We describe the reservoir by the n traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t = (1-p)\,O(u_t)\,r_{t-1} + p\,b. \tag{B1}$$

Here b represents the reset state, and the inputs u_t are independent and identically distributed. A stored signal loses a fraction $2p$ of its variance per step, so it lasts about $t_{\mathrm{mem}} = 1/(2p)$ steps. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A = \mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs scramble.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^{\mathsf{T}}] = X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2 > 0$ with $\mu = (1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s = 1 - \frac{\|A\|_F^2}{n} \tag{B2}$$

says how strongly one input turns the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-A)x\|^2 = s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, and the first condition makes it positive. If the inputs $u = \pm 1$ rotate by $O = e^{u\theta G}$ with a fixed generator G, whose eigenvalues are $i\lambda_1, \dots, i\lambda_n$, the average rotation is $\cosh \theta G$ and $s = \frac{1}{n}\sum_j \sin^2 \theta\lambda_j$, which grows as θ^2 for small turns.

Mixing law, precise form. Assume conditions (1)–(3) and take any fixed linear readout. Pick a stored function with probability equal to its share of the readout's capacity, and let D be its order. As $p \to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

Equivalently, products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

B.2 The idea: a race

A stored signal is spread evenly over all directions. In each step the current input turns a fraction s of it away from the average rotation, which adds one input to that part, and the reset removes a fraction $1/t_{\mathrm{mem}}$. On average $1/s$ steps pass before the next input joins a product, and meanwhile it loses a fraction $1/(s\,t_{\mathrm{mem}})$ of its weight, so products of more than K inputs carry about

$$\Big(1 - \frac{1}{s\,t_{\mathrm{mem}}}\Big)^K \approx e^{-K/(s\,t_{\mathrm{mem}})},$$

an exponential law with mean $s\,t_{\mathrm{mem}}$. The proof makes this bookkeeping exact.

B.3 Proof

The proof works with covariance matrices. Appendix A.4 splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (A3): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^{\mathsf{T}}$, the full step $\mathcal{T}X = (1-p)^2\,\mathbb{E}[OXO^{\mathsf{T}}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1-(1-p)A]^{-1}b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2 v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged. Since they cannot enlarge a covariance, the identity is a simple fixed point with the projector $X \mapsto (\mathrm{Tr}\,X/n)\,1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1}V = \frac{pv}{2n}\,1 + O(p^2). \tag{B3}$$

Step 2: few inputs carry almost nothing. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1}V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout, although the shares of all orders add up to its rank. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F}G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^{\mathsf{T}} = (1-\mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \mathrm{Tr}[(1 - A^{\mathsf{T}}A)X], \qquad \ell(1) = ns, \quad \text{(B4)}$$

since $\ell(\mathcal{F}_0 X) = \mathrm{Tr}\,\mathcal{K}_0 X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\,p + O(p^2)\,: \qquad \text{(B5)}$$

each step up in order costs a fraction $2p/s = 1/(s\,t_{\mathrm{mem}})$ of the weight, as in the race. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2\cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0' 1) = -2n$, and dividing by $\ell(1) = ns$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\lambda p k}$. Summing the recursion gives $\sum_k z^k G_k = z(1 - z\mathcal{F})^{-1}G_1$, and as $z = e^{-\lambda p}$ approaches one, only the eigenvalue of Eq. (B5) makes the inverse large. With $\ell(G_1) = p^2 v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k\geq 1} e^{-\lambda p k} G_k = \frac{pv}{2n}\,\frac{\beta}{\lambda+\beta}\,1 + O(p^2). \qquad \text{(B6)}$$

Divided by Eq. (B3), this is $\beta/(\lambda+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\mathrm{mem}} = 2pD$ is exponential with mean s. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k k G_k = (1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1)$.

Two technical points. The eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point of $\mathcal{F}_0$ satisfies $\mathcal{T}_0 X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1-\mathcal{F})^{-1} = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (B5) makes the sum diverge.

B.4 When the law needs one slow direction

Condition (2) matters. Let a classical flag, set by the reset state and never changed, select one of two qubit rotations about Z, by $\pm\pi/6$ or by $\pm\pi/3$, and let the reset state put the flag in either value with probability $\frac{1}{2}$ and the qubit along $+X$ in both. Each sector obeys the law with its own mixing fraction, $1/4$ and $3/4$, and a single Y readout sees their mixture, weighted by the variance each sector stores. The gentler sector stores more, so as $p \to 0$ products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $w\,e^{-4x} + (1-w)\,e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce. One observable then shows two order scales.

B.5 Physical channels without a reset

A physical channel does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general channels. Appendix D uses this version, and readers interested in the reset model can skip it.

Setting. One step is $r_t = A_u(\epsilon)r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$.

(ii) The mean step forgets at $\epsilon = 0$: $\mathbb{E}A_u(0)$ has spectral radius below one.

(iii) The channel is mean-square stable for small $\epsilon > 0$.

(iv) One covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one. At $\epsilon = 0$ its right eigenvector is a positive matrix Q,

and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$.

(v) The input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/n$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of a shared channel with opposite phases differently in each step and spread a stored signal over all of them, so that one covariance direction stays slow. With several slow directions, each mixes at its own rate, as in the example above. Conditions (ii) and (iii) say that the reservoir forgets its starting state on average and in mean square, (iv) that the random steps spread a stored signal over the slow modes faster than it leaks, and (v) that the surviving direction is written.

Mixing law without a reset. Under (i)–(v) the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory time $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon\,\frac{\ell(V_*)}{\kappa}\,Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. For the memory of a readout with several features we also need Q to have full rank on what the readout measures, which holds trivially for one observable. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1 - \mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1 - \mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1 - \mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1 - (\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

The spin reservoirs. For three spins the certificates of Appendix D check these conditions. For five spins we check the law instead of the conditions: Eq. (A3) gives the order distribution exactly, and it follows the law (below).

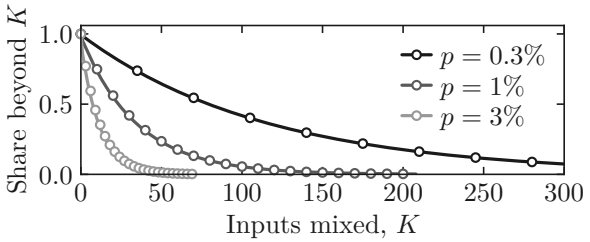


Figure 5: **The mixing law.** Share of the capacity carried by products of more than K inputs, for one qubit that is reset with a small probability p in each step. Lines: exact, circles: the exponential law of Theorem 1.

B.6 Numerical check

Figure 5 compares the law with exact moments. Beyond the reservoir shown, we tested 24 reservoirs and readouts in dimensions two to four: two families of three random unitaries, two reset states (a basis state and a random pure state) and two readouts (all coordinates or one random direction). The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. The five-spin reservoir follows the law as well (Fig. 3c): at the largest memories computed, $s\,t_{\mathrm{mem}} \approx 20$, products of more than K inputs carry the share $e^{-K/(s\,t_{\mathrm{mem}})}$ of the capacity to within 0.14.

C One qubit in closed form

This appendix solves the qubit of Sec. 3.3 exactly. At a fixed turn per input the number of inputs mixed grows as the memory time. A turn that shrinks as the square root of the relaxation rate keeps it finite, and the stored signal is largest at an intermediate turn.

The model. The qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|{+x}\rangle\langle{-x}|$, and the input $u = \pm 1$ turns it about Z by $u\theta$ per unit time. Each input lasts one unit of time, and the

qubit is read along Y. In the Bloch plane,

$$\dot{x} = -\epsilon(x-1) - u\theta y, \qquad \dot{y} = u\theta x - \tfrac{1}{2}\epsilon y. \quad \text{(C1)}$$

Over one step this is a linear map of (x, y) plus an offset. For $u = +1$ the map has the form $\left(\begin{smallmatrix} a & -b \\ b & d \end{smallmatrix}\right)$ and the offset is (f, g). For $u = -1$ the signs of b and g flip. The input-averaged step is therefore diagonal, the stationary mean lies on the X axis at $m = f/(1-a)$, and each input writes $u\,v$, with $v = bm + g$, along the measured axis Y.

How the orders arise. The averaged step keeps the two axes apart, while the input-dependent part swaps them. A signal written into Y must pass through X and back to pick up more inputs, so Y carries only odd orders. By the recursion (A3), covariance passes from Y to X with the factor $b^2/(1-a^2)$ and back with $b^2/(1-d^2)$. Call the product of the two factors χ. Order $2j+1$ then carries the share $(1-\chi)\chi^j$ of the capacity of Y, and

$$\mathbb{E}D = \frac{1+\chi}{1-\chi}, \qquad \operatorname{Var} Y = \frac{v^2}{(1-d^2)(1-\chi)}. \quad \text{(C2)}$$

Single past inputs carry the share $d^2(1-\chi)$.

Fixed turn. Let the relaxation slow down, $\epsilon \to 0$, at a fixed angle θ with $\sin\theta \neq 0$. Then $1-\chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$: the order grows as $1/\epsilon$, ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes, even for exact expectation values.

Gentle writing. If instead θ^2/ϵ approaches a constant ν, then to first order $a = 1-(1+\nu/2)\epsilon$ and $d = 1-(1+\nu)\epsilon/2$, with $b \approx \theta$ and $f \approx \epsilon$, and as $\epsilon \to 0$

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \qquad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \quad \text{(C3)}$$

The mean order stays finite, and single past inputs keep the share $1-\chi$. Their recall beyond delay τ is the fraction $d^{2\tau}$ of the total, so the range that holds 90% of it grows as $1/\epsilon$: $\epsilon\tau_{90} \to \ln 10/(1+\nu)$.

The best turn. The signal of Eq. (C3) is largest at $\nu = 2/\sqrt{3}$ (Fig. 3d). For $\nu \gg 1$ it vanishes as $4/(3\nu)$, because later inputs scramble earlier ones. For $\nu \ll 1$ it vanishes as ν, because each input writes too little. Among power laws $\theta \propto \epsilon^\alpha$, only $\alpha = 1/2$ keeps a finite signal: the turn must fall as the square root of the relaxation rate, that is, as one over the square root of the memory time. Direct propagation of the density matrix reproduces all these expressions.

D Three spins at strong loss

What we prove. Take three spins with one shared channel of opposite phases, Eq. (D1), the four Hamiltonians H1–H4 of Table 3, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- At a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the signal in X_1 falls as $1/\gamma$ (boxes of half-width 1/32).
- With the amplitude scaled as $8/\sqrt{\gamma}$, single past inputs keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their recall grows in proportion to γ (boxes of half-width 1/512).

Idea. At strong loss, one step of the full master equation is a rotation of the three dark states, up to corrections of order $\epsilon = 1/\gamma$. At a fixed amplitude the two signs of the input rotate the dark states differently. The stored covariance then has a single slow direction, and the mixing law without a reset applies (Appendix B.5). With gentle writing both signs give almost the same rotation. The signal is written on the scale of the leak and collects in six slow directions, which we follow explicitly. Both arguments end in finitely many strict inequalities, which a computer checks with rigorous error bounds.

Model and computations. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2}\sum_i Z_i + \tfrac{3+\eta u}{4}\sum_i X_i \quad \text{(D1)}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ (fixed amplitude) or $\eta = 8/\sqrt{\gamma}$ (gentle writing): the field of gentle writing (Appendix A.3) with signs in place of uniform inputs, with the same mean

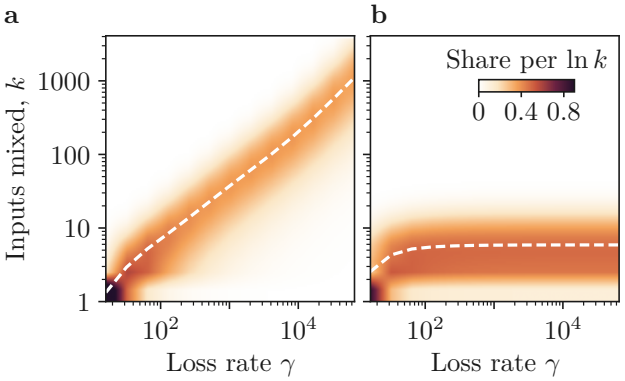


Figure 6: **Three spins, both regimes.** Share of the capacity of X_1 carried by products of k inputs (colour) against the loss rate, for one shared channel with opposite phases (Hamiltonian H3). Dashed: the mean. At a fixed input strength the number of inputs mixed grows with the loss (a), while with gentle writing it stays small (b).

Table 3: Strong-loss limits for X_1 in the four three-spin reservoirs. Fixed amplitude: rate of the exponential law of the order divided by γ (certified). Gentle writing: share of single past inputs and range of their recall divided by γ, evaluated at $\gamma = 2^{16}$.

	rate	single-input share	range/γ
H1	78.6	0.81	0.50
H2	91.7	0.29	0.18
H3	61.4	0.08	0.24
H4	132.5	0.14	0.50

and the same largest change $\eta/4$. The couplings of the four Hamiltonians H1–H4 are drawn from fixed seeds. For each loss rate we exponentiate the full generator for both signs, solve the stationary moment equations and evaluate Eq. (A3) for X_1 until 99.99% of its memory is accounted for. The range τ_{90} is the smallest delay beyond which less than a tenth of the recall of single inputs remains.

Both regimes. Figure 6 shows the two regimes of the box for H3. Gentler writing trades signal for readability, as for the qubit: at $\eta_0 = 8$ single inputs keep between 8% and 81% of the capacity of X_1 for the four Hamiltonians (Table 3), and for H3 this share rises to 67% at $\eta_0 = 2$, with a signal 25 times smaller.

One step at strong loss. The jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$ leaves three dark states: the state with no excitation and the two single-excitation states with amplitudes $(0, 1, -1)/\sqrt{2}$ and $(2, 1, 1)/\sqrt{6}$, which are orthogonal to the phases of the channel. It empties every other state, so at unit loss rate only the nine operators on the dark states do not decay. Let R embed these nine coordinates into the full operator space and $R^\flat$ project back along the decaying operators. Expanding one step in ϵ gives

$$E_u = R\, e^{\Delta t\, R^\flat \mathcal{H}_u R} R^\flat + \epsilon Z_u + O(\epsilon^2), \qquad \text{(D2)}$$

where $\mathcal{H}_u$ is the commutator with the Hamiltonian of input u. The first term rotates the dark states. The correction Z_u collects the brief excursions into bright states: one at the start of the step, one at its end, and the leak of Eq. (A2) in between.

Fixed amplitude. At $\eta = 1$ the dark rotations O_u of the two signs differ, and the computer checks the conditions of the mixing law without a reset on the eight traceless dark coordinates. The average rotation forgets. The identity is the only covariance that the random rotations leave unchanged. Three constants from the first-order terms of Eq. (D2) are positive: the strength with which an input writes, the rate κ at which the slow covariance leaks, and the mixing fraction $s_0 = 1 - \|\mathbb{E} O_u\|_F^2/8$. The law then gives an exponential distribution of the order divided by γ, with rate κ/s_0 (Table 3), and the variance of X_1 falls as ϵ. A change r of the couplings moves each dark rotation by at most $3r$, which lets the checks cover the boxes of half-width 1/32.

Gentle writing. At $\eta = 8\sqrt{\epsilon}$ both signs give the same leading rotation E, generated by the static part of the dark Hamiltonian. The input enters at order $\sqrt{\epsilon}$ and the leak at order ϵ. Expanding the step to this order gives the mean step $E + \epsilon E_1$ and the input-dependent part $u\sqrt{\epsilon}\,\Delta_1$.

Where the signal goes. The rotation E turns each of the three planes of coherence between the dark energy levels by its own angle, set by an energy gap, and the six phases of these turns differ from each other and from one. So E leaves only the populations unchanged, and on them E_1 generates a slow population dynamics with a unique stationary state m_*. An input therefore writes $\pm\sqrt{\epsilon}\, v$ with $v = \Delta_1 m_*$, which lies in the coherence planes: a new input first turns a population into a coherence.

Six slow directions. The covariances that E leaves unchanged form a six-dimensional space: the three covariances of the two population coordinates and one isotropic covariance in each

coherence plane. Over $1/\epsilon$ steps a covariance in this space changes under two generators, both certified stable. K_B describes how the mean step lets a stored signal decay, and K_T adds the random signs of later inputs, which turn a stored signal into products. With P_6 the projection onto the six directions, the single-input part of the stored covariance and the full covariance approach $W_* = -K_B^{-1} P_6(vv^\mathsf{T})$ and $\Sigma_* = -K_T^{-1} P_6(vv^\mathsf{T})$.

Limits. With a the coordinates of X_1, the share of single past inputs in the memory of X_1 tends to $a^\mathsf{T} W_* a / a^\mathsf{T} \Sigma_* a$. After t/ϵ steps the remaining single-input recall is the fraction $a^\mathsf{T} e^{tK_B} W_* a / a^\mathsf{T} W_* a$, which crosses 0.1 once, and the crossing is the range divided by γ. The certificates show $a^\mathsf{T} W_* a > 0$, and since $\Sigma_* \succeq W_*$ the share is positive. In the boxes of half-width $1/512$ the six angles stay separated and both generators stay stable, so the limits stay positive.

Why coherence carries the memory. Returning bright population to the dark subspace after every step leaves the leading terms of the scaled step unchanged, so it keeps the same limits. Dephasing in the dark energy basis, by contrast, removes the coherence planes into which v writes. Restricted to populations, the two signs then differ only at order $\epsilon^{3/2}$, because the response of a population at order $\sqrt{\epsilon}$ is a coherence, while the populations still follow the slow population dynamics. A new input thus writes a vector of size $\epsilon^{3/2}$, the slow dynamics sum its square over $1/\epsilon$ steps, and the stored signal is of order ϵ^2: the collapse as $1/\gamma^2$ of Sec. 3.3. Numerically, applying the loss-only projector $\mathcal{P}$ after every step, with or without dephasing in the energy basis of the projected Hamiltonian, the variance of X_1 stays between 10^{-4} and 2×10^{-3} from $\gamma = 2^8$ to 2^{18} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased. At a fixed input amplitude it falls as $1/\gamma$.

What the computer checks. The attached archive `spin_certificates.zip` contains the generators, the exact couplings and the interval programs. They enclose every quantity above in intervals with rigorous rounding (ball arithmetic of Arb, python-flint 0.9.0) and check each strict inequality of this appendix at 256 bits, and at 192 bits they reproduce the archived enclosures. A second program, sharing no routine with them, builds the full finite-rate maps and extrapolates to infinite loss. It agrees with the certified rates and limiting recalls to better than one part in a million.

E Numerical details and supporting checks

This appendix backs the main text: each paragraph names the section it supports, says what we checked and gives what is needed to reproduce it.

Recall and memory (Sec. 3.1). Figures smooth recall curves for display only: each delay is averaged with its neighbours, with weights that fall linearly to zero at a fifth of the delay, and no sum uses the smoothed values. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^6 in powers of two and interpolated by shifting each recall profile along the delay. Figure 2c stops one and two shared channels at $\gamma = 64$, where their recall at the full amplitude starts to reach past the 300 delays we sum. Summed over all delays with exact moments (Appendix A.4), their memory keeps rising up to $\gamma = 256$. With gentle writing, delays beyond 300 add about 1% to the memory of opposite phases.

Dark states (Sec. 3.1). Both tests act on the 100 dark coordinates. For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the dark coordinates α, evolve α under Eq. (A2) and record the same 45 features of the full state $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$. The readout is trained on them exactly as for the full reservoir, and the model has no fitted parameter (Fig. 7a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$. The control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is measured at delays 1 to 100 with 800 training, 200 validation and 800 test inputs (Fig. 7b).

The leak out of the dark states falls exactly as $1/\gamma$: for one shared channel $\gamma\bar{\kappa}$ approaches $\kappa_\mathrm{c} = 2.945$ from $\gamma = 512$ on, so t_f grows in proportion to γ, while it levels off near 56 steps with opposite phases and near 110 with pairwise loss. The second-order calculation of Eq. (A2) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three,

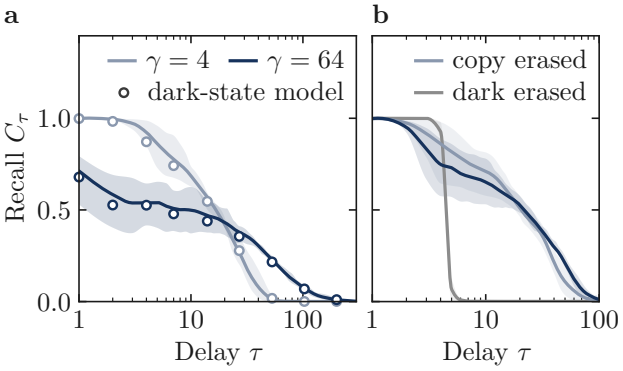


Figure 7: **Dark states hold the memory.** One shared channel. **a** The dark states alone (circles), with no fitted parameter, reproduce the recall of the full reservoir (lines). **b** Erasing the dark states after every step destroys the memory, while erasing a rotated copy of the same size leaves most of it (dark line: intact reservoir, $\gamma = 16$).

four and six spins: the size of the reservoir sets only the strength of the leak.

Time to forget and phases (Sec. 3.2). We compute t_f at $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$, for γ from 2^{-8} to 2^6 with local loss and from 2^{-2} to 2^{12} with opposite phases. The collapses of Fig. 3a do not depend on how the time to forget is defined: with the slowest mode in place of the mean rate they hold to within 11% wherever forgetting takes more than four steps, and they hold as well with the input on one spin. Weakening the input moves the best loss rate: on one reservoir with exact moments, the memory peak of local loss moves from $\gamma \approx 1/16$ at $\eta = 1$ to $1/64$ at $\eta = \frac{1}{8}$, and the best memory rises from 16 to 39 units.

The write per input of Fig. 3b is $2\|R^\flat[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \dots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. Along the phases it falls as $1/\gamma$, to 0.003 at $\gamma = 128$, while across them it falls only slowly, to 0.18. Opposite phases are the equal-phase channel with the field on spins 4 and 5 reversed, up to a relabelling of the random couplings (a turn of these spins by π about Z).

Orders and capacities (Secs. 3.2 and 3.3). The order distribution of Eq. (A3) uses six-point Gauss–Legendre quadrature for the expectations over the input and solves the equations $X - \bar{A}X\bar{A}^\mathsf{T} = C$ in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The linear recall follows from the same moments, $C_\tau = c_\tau^\mathsf{T}\Sigma_y^{-1}c_\tau/\mathrm{Var}\,u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance, and reproduces the simulated memories to a median of 2%. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_mem, Q and ℓ. Loss rates run in powers of four over the ranges of Fig. 3c, and with gentle writing to 16384 and 1/4096, on four reservoirs, two at the largest memory times. With local loss the mean order follows the law as well, within 2.2 of $s\,t_\mathrm{mem}$ up to $s\,t_\mathrm{mem} = 18$.

Measurement (Sec. 3.4). We measure the signal by the variance of a feature over the test inputs, averaged over the features. Gentle writing makes it fainter: with opposite phases it falls from 8×10^{-5} at the full amplitude to 5×10^{-6}, against 7×10^{-5} with local loss and gentle writing at $\gamma = 1/32$. Restarting each run from $|00000\rangle$, as the replay does, raises these to 4×10^{-5} and 1.2×10^{-4}. At the full amplitude the signal of the equal-phase channel falls from 2×10^{-4} at $\gamma = 1$ to 4×10^{-7} at $\gamma = 64$, which is why this channel loses memory as the loss grows once measurements are counted.

The replayed range τ_{90} grows with the clock, though more slowly: along the gentle line of opposite phases from about 30 steps at $\gamma = 16$ to 130 at $\gamma = 256$. The budget sets the best loss rate: at 10^{10} preparations times replayed steps the optimum of the equal-phase channel moves from $\gamma = 1$ to about 16, with 17 units, while opposite phases with gentle writing peak near $\gamma = 64$ with 25. Below the crossings the full amplitude wins: at 10^4 preparations it keeps 11 units with local loss and 8 with opposite phases, against 9 and 3 with gentle writing. With the input on one spin, which writes more gently by itself, the crossings move to about 3×10^5 and 4×10^6 preparations. With the full history, eight times more training inputs change the sampled memory by less than 1.3 units.

For the matched clock of Fig. 4b we find, for every shape and reservoir, the loss rate at which $t_\mathrm{leak} = 32$ steps, at weak loss without protected operators and at strong loss with them, and run the reservoir there with $\eta = \sqrt{4/32}$. The write is $-i[H_\mathrm{in}, \rho_*]$ with $H_\mathrm{in} = h\sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow

modes: the protected ones, or without protected operators those decaying at most four times faster than the slowest. The memory that survives 10^6 preparations rises with the write, from 0.3 units for the shared channels to 17 for local loss, and power laws fitted within the shapes without and with dark states (dotted in Fig. 4b) have exponents 2.0, as linear accumulation of the write predicts, and 1.5.

Ordinary noise (Sec. 4). Adding local loss $\sqrt{\gamma_{\rm loc}}\,\sigma_i^-$ to a shared channel gives $\bar{\kappa} = \kappa_{\rm c}/\gamma + \kappa_{\rm l}\gamma_{\rm loc}$, where to first order $\kappa_{\rm l}$ is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$. Local dephasing $\sqrt{\gamma_\phi}\,Z_i$ adds $\kappa_\phi\gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$, and the full spectra reproduce both coefficients to within 0.2%. A time to forget of at least T steps, $\kappa_{\rm c}/\gamma + \kappa_{\rm l}\gamma_{\rm loc} \leq 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\rm loc} \geq 4\kappa_{\rm c}\kappa_{\rm l}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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